

A Spreading-Activation Model of Causal Reasoning

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Abstract

Human causal reasoning can flow in two directions: *prediction* involves inferring effects from causes, and *diagnosis* involves inferring causes from effects. Empirically, the direction of reasoning is associated with various psychological asymmetries. Here, we present a mechanistic computational cognitive model of causal reasoning that can account for these phenomena. Specifically, we propose that causal reasoning may be viewed as a memory search process over causal knowledge implemented through elementary mechanisms of spreading activation, lateral inhibition, and competitive retrieval, and that empirical asymmetries associated with the direction of causal reasoning emerge from divergent search priorities. We then demonstrate that our model (*BLISA*) generates testable behavioural predictions through a simulation study in which the model generates distinct individual-level choice and response-time patterns for predictive and diagnostic queries on various causal structures.

Keywords: causal reasoning; spreading activation; lateral inhibition; cognitive architecture

Introduction

Playing an electric guitar with an unshielded instrument cable tends to make the amplifier noisier. This knowledge can be deployed in two directions: A musician can *diagnose* a noisy amp, attributing it to an unshielded cable, or the musician can *predict* that using an unshielded cable will generate a desired noisy amp effect. The former infers cause from effect (backward direction), and the latter infers effect from cause (forward direction). Importantly, these two directions of causal inference present several empirical asymmetries.

First, human reasoners exhibit a preference for, and greater fluency with, the predictive direction over the diagnostic direction, possibly because the former is less cognitively demanding. Notably, across a seminal series of studies, Tversky and Kahneman (1980) find that people judge predictive inferences as more certain than diagnoses, even when the probabilities are equal in both directions. Furthermore, when given the opportunity to question a faulty causal explanation (e.g. X occurred because of Y), people more readily challenge the effect than the cause, generating more alternative effects than alternative causes.

Moreover, human reasoners tend to neglect alternative causes when reasoning in the predictive mode, but not in the diagnostic mode. For instance, Fernbach et al. (2011) manipulate whether or not an alternative cause for an effect is presented prior to a causal judgment task. Some groups

are not explicitly reminded about the alternative cause with their test question (experiment 1, 2), while others are (experiment 3). When diagnosing the likelihood of a cause given its effect, the participants' judgments are found to account for the alternative cause even without reminder. However, when predicting the likelihood of an effect given one of its causes, participants' judgments fail to account for the alternative in the no-reminder case.

Finally, people tend to look for only one cause in diagnosis tasks, even when many alternative causes may be involved. This phenomenon is known as the *single-cause fallacy*. In medicine, when one diagnosis is identified, clinicians become less likely to look for another (Redelmeier & Shafir, 2023). When people are aware of many potential causes, observing just one tends to lower expectations about others (Kelley, 1973). And, in causal learning, "blocking effects" describe competition among causes: people associate effects with the first-learned cause rather than later-learned causes, even when both independently predict the outcome (Waldmann & Holyoak, 1992), whereas effects associated with a common cause do not exhibit such competition (Waldmann, 2000).

These asymmetries are notable in that they are not readily explained by normative models of causal inference and sometimes even contradict such models outright (see, e.g., Fernbach et al., 2011; Rottman and Hastie, 2014; Tversky and Kahneman, 1980). Here, we present a computational cognitive model of causal reasoning that may provide mechanistic explanations for these empirical asymmetries using the elementary cognitive mechanisms of spreading activation, lateral inhibition, and competitive retrieval. Furthermore, to demonstrate that our model generates testable empirical predictions, we propose a novel experimental paradigm and report results from a simulation study. Our model is formulated using the Clarion cognitive architecture (Sun, 2016) and is capable of simulating individual-level choices and response times.

Model

Like normative models of probabilistic causal reasoning, we assume that causal knowledge is organized in the form of *causal graphs*, i.e., networks where nodes represent events and edges represent direct causal dependencies (see, e.g., Danks, 2014; Gopnik et al., 2004; Pearl, 1986). However, unlike such normative models, we conceptualize the response process associated with causal queries as a kind of memory

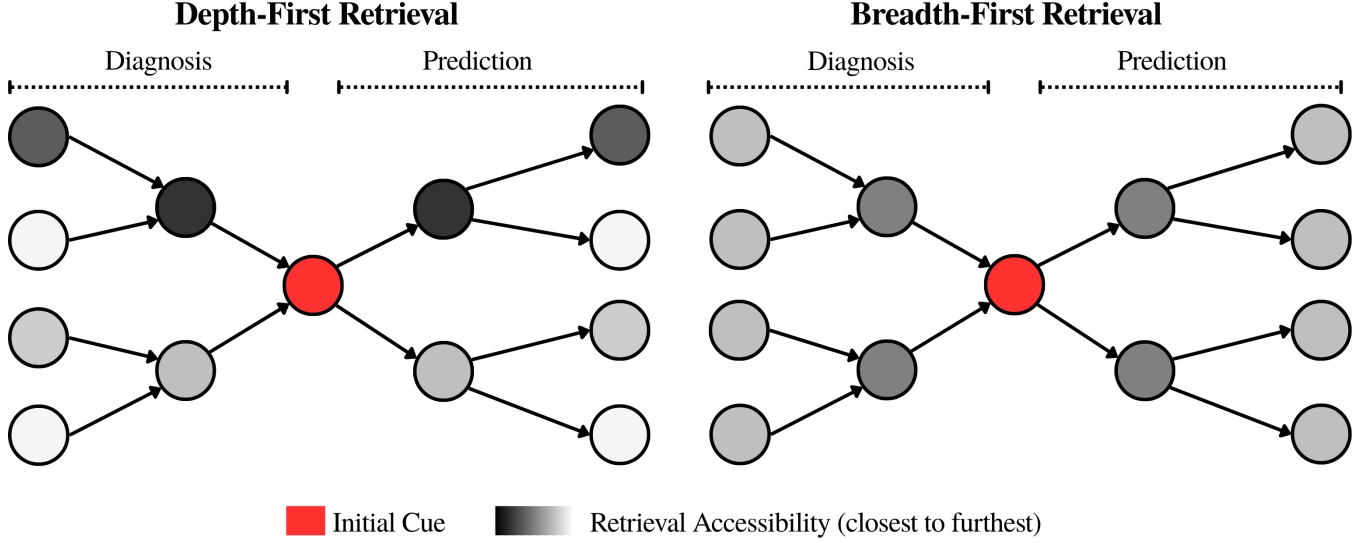


Figure 1: Example retrieval paths for each preference pattern and causal direction. The initial cue is the event given as evidence for an inference query. Retrieval accessibility indicates the retrieval priority of events in either direction of inference according to each pattern. More accessible (darker) nodes are likely to be retrieved earlier than less accessible (lighter) nodes.

search over causal networks. In particular, we hypothesize that responses to diagnostic (predictive) causal queries are generated via repeated retrievals of causes (effects) associated with events in working memory. Such sequences of retrievals effectively construct causal narratives, working backwards (forwards) from a given effect (cause), in a manner comparable to the elaboration processes posited by mental model theory (Johnson-Laird and Byrne, 1991). Thus, the response process terminates affirmatively if the queried cause (effect) is retrieved and negatively upon the first failure to retrieve any cause (effect).

Such a model can capture asymmetries in causal reasoning via its search dynamics. To start, neglect for alternative causes in predictive reasoning (e.g., Fernbach et al., 2011) is readily captured by the absence of alternative causes in working memory. Capturing a preference for single-cause explanations, as suggested by the prevalence of the single-cause fallacy (e.g., Kelley, 1973; Redelmeier and Shafir, 2023), requires stronger assumptions about how search dynamics differ between the two types of inference.

Figure 1 illustrates two retrieval patterns that may be used to elaborate a causal narrative within the model. The breadth-first pattern prioritizes retrieval of direct dependencies whereas the depth-first pattern prioritizes retrieval of dependencies along a single chain of causes. Thus, the model may capture a preference for single-cause explanations if the depth-first pattern characterizes processing during diagnostic inference whereas the breadth-first pattern characterizes processing during predictive inference. Moreover, this configuration implies a greater cognitive cost and difficulty for diagnostic inference as compared with predictive inference

(as suggested by, e.g., Tversky and Kahneman, 1980), since a target is less likely to be on a preferred search path in the former case whenever the causal structure includes branching in the diagnostic direction.

In light of the above, our model is called *BLISA* (Bidirectional causal reasoning using Lateral Inhibition in a Spreading Activation process). Structurally, BLISA consists of a working memory, a body of associative rules encoding its causal knowledge, and a declarative retrieval mechanism. Mechanistically, these components correspond to the top level of the Non-Action Centered Subsystem (NACS) in the Clarion cognitive architecture (see Sun, 2016). The model also exhibits some novel capabilities relative to the standard configuration of the NACS. Notably, the model is capable of selectively propagating activations in either direction through its associative rules. Furthermore, the model is equipped with a lateral inhibition mechanism which inhibits causes that share effects with events already encoded in working memory, enabling the model to implement the depth-first retrieval pattern in diagnostic inference.

Formally, the model encodes causal knowledge in the form of a set C of chunks and a set $\mathcal{R}$ of Clarion-style *associative rules*. Events are represented by chunks and direct causal dependencies by associative rules of the form $i \rightarrow j$ linking a cause i to an effect j . For $i, j \in C$ and $r \in \mathcal{R}$, the model maintains a rule antecedent matrix $A = (a_{ri})$ and a rule consequent matrix $C = (c_{ir})$. A direct causal dependency r of the form $i \rightarrow j$ is therefore initialized by setting a_{rk} and c_{kr} as follows.

$$a_{rk} = \begin{cases} 1 & \text{for } k = i \\ 0 & \text{for } k \neq i \end{cases} \quad c_{kr} = \begin{cases} 1 & \text{for } k = j \\ 0 & \text{for } k \neq j \end{cases}$$

Additionally, the model maintains a shared effect matrix $H = (h_{ij})$ supporting its lateral inhibition processes. The matrix H is derived from the causal adjacency matrix $W = CA$ such that $H = W^T W$. Thus, w_{ij} and h_{ij} are computed as follows.

$$w_{ij} = \sum_{r \in \mathcal{R}} c_{ir} a_{rj} \quad h_{ij} = \sum_{k \in \mathcal{C}} w_{ki} w_{kj}$$

Each chunk $j \in \mathcal{C}$ is initialized with a working memory activation, $x_j(0)$, such that $x_j(0) = 1$ if j represents the initial event of the current causal query (i.e., the effect in a diagnostic query or the initial cause in a predictive query) and $x_j(0) = 0$ otherwise. The model proceeds by updating each chunk's working memory state ($x_j(t)$) through a sequence of retrievals ($m(t)$) as follows.

$$x_j(t+1) = \begin{cases} 1 & \text{for } j = m(t) \\ x_j(t) & \text{for } j \neq m(t) \end{cases}$$

Retrieval is carried out by a standard competitive process (McFadden, 1974) weighing activations between candidates. The chunk set $\mathcal{C}$ includes a special nil chunk with no causal dependencies representing retrieval failure. Competitive retrieval is implemented by adding Gaussian noise $\varepsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$ to chunk activations and selecting the chunk with the greatest noisy activation:

$$m(t) = \arg \max_{i \in \mathcal{C}} \{s_i(t) + \varepsilon_i(t)\}$$

Chunks with higher activations are more likely to be retrieved, but noise allows occasional intrusions of other options. The standard deviation of noise, σ , controls the likelihood of such intrusions.

The activation of a chunk $i \in \mathcal{C}$ is defined by the following formula, where $s_i^{\mathcal{P}}(t)$, $s_i^{\mathcal{D}}(t)$, $s_i^{\mathcal{H}}(t)$ are, respectively, the predictive signal, the diagnostic signal, and the lateral-inhibitory signal for i , and α_p , α_d , α_h , and α_x represent meta-cognitive weights for each input.

$$\begin{aligned} s_i(t) &= \alpha_p s_i^{\mathcal{P}}(t) + \alpha_d s_i^{\mathcal{D}}(t) + \alpha_h s_i^{\mathcal{H}}(t) + \alpha_x x_i(t) \\ s_i^{\mathcal{P}}(t) &= \max \{c_{ir} a_{rj} x_j(t) : j \in \mathcal{C} \wedge r \in \mathcal{R}\} \\ s_i^{\mathcal{D}}(t) &= \max \{a_{ri} c_{jr} x_j(t) : j \in \mathcal{C} \wedge r \in \mathcal{R}\} \\ s_i^{\mathcal{H}}(t) &= - \sum_{j \in \mathcal{C}} h_{ij} x_j(t) \end{aligned}$$

Here, the predictive signal $s_i^{\mathcal{P}}(t)$ implements the standard equation for Clarion-style associative rules of the form $j \rightarrow i$, the diagnostic signal $s_i^{\mathcal{D}}$ inverts this computation for rules of similar form, and the lateral-inhibitory signal negatively influences siblings of chunks available in working memory.

Finally, the model's meta-cognitive weights provide an interface to control the memory search process. To reduce the number of free parameters and to capture the processing patterns discussed above, we assume $\alpha_p = 1$ and $\alpha_d = \alpha_h = 0$ in predictive inference, and in diagnostic inference, we assume $\alpha_p = 0$ and $\alpha_d = \alpha_h = 1$. Furthermore, we set $\alpha_x = -1$ to prevent retrieval of chunks already in working memory. Within the Clarion architecture, these parameter settings may be controlled by the Action-Centered or Metacognitive Subsystems (Sun, 2016). However, our present simulations do not instantiate these subsystems for simplicity of implementation.

Simulation Study

To explore possible empirical predictions implied by our model, we propose a novel experimental paradigm designed to elicit behavioural asymmetries in causal reasoning, and we present a simulation study that runs our model on this task.

Method

The proposed paradigm requires reasoning about causal networks with topologies designed to elicit behavioural asymmetries between the two directions of causal inference. The main task involves using provided causal knowledge to infer whether the occurrence of a given event suggests the occurrence of a *query event*. These questions can prompt predictive reasoning (cause to effect) or diagnostic reasoning (effect to cause).

Design The proposed experiment employs a 3 (Causal Scenario: CHAIN, COMMON CAUSES, COMMON EFFECTS) $\times$ 2 (Direction: PREDICTION, DIAGNOSIS) $\times$ 2 (Length: SHORT, FULL) design. The Causal Scenario factor varies the topology of the underlying causal network. CHAIN serves as the control scenario as it contains no branching. The COMMON CAUSES condition has branching effects, and the COMMON EFFECTS condition has branching causes. The Direction factor varies the direction of causal inference. Thus, if the scenario implies that A is a (possibly distal) cause of B , then, in prediction, B is the query and A the given event and, in diagnosis, these roles are reversed. Finally, the Length factor varies the distance between the evidence and the query in the test question in terms of the number of intervening causes along the relevant causal path.

Measures The primary dependent measures are response time and response correctness, both of which may be interpreted as indices of processing difficulty.

Materials We introduce the case of a self-driving golf cart crash as a working example for experimental causal scenarios. In all experimental scenarios the primary narrative consists of a chain of three focal events: a malware infection (A) causes impaired self-driving (B), which causes a crash (C). However, the three scenarios contain different auxiliary causal dependencies, as depicted in Figure 2.

- I. CHAIN: Single causal chain from malware to impaired driving to crash without any branching.
- II. COMMON CAUSES: Malware and impaired driving each have additional effects: changes to font settings and strange (but non-dangerous) manoeuvres, respectively.
- III. COMMON EFFECTS: In addition to the causal chain from malware to the crash, there is another chain from a missed maintenance session to brake failure to crash.

Thus, in Condition I, participants may be presented with the following vignette:

Sheraton Grand Rio Hotel & Resort staff study the case of a self-driving golf cart crashing into an outdoor steel

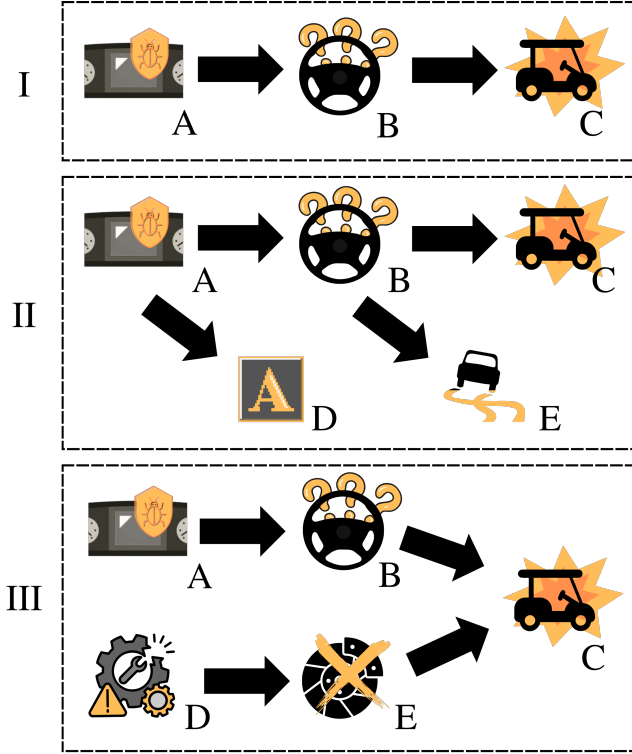


Figure 2: Depiction of vignettes for each experimental causal scenario.

beam. They were told that crashes can mainly be attributed to impaired driving. The resort’s golf carts are self-driven via artificial intelligence software which is generally free from impairment except when infected by malware, in which case the malware leads to impaired driving.

Vignettes for Conditions II and III describe the same focal causal chain but also include the relevant additional causal dependencies.

Procedure Participants start by reading vignettes about potential causes for different scenarios. Following this, participants respond to a series of comprehension questions and then a series of experimental trials. Comprehension questions ask whether participants understand every causal relationship presented in the vignette. An example question in the *CHAIN* scenario could be “Does malware cause impaired driving?” for which the correct answer is yes. Participants who fail comprehension questions are excluded to ensure that response variation cannot be attributed to differences in causal knowledge.

Experimental trials consist of causal inference questions testing participants’ causal reasoning and are in the conditional form “if X occurred, could Y have occurred?” Here, the experimenter provides X as evidence and queries Y. Depending on the Direction condition, the experimental trial prompts *PREDICTION* (X occurs before Y) or *DIAGNOSIS* (X occurs after

Y). The Length factor varies between one (*SHORT*) and two (*FULL*) intervening dependencies. Thus, *SHORT* trials query the dependency relationships between B and C, and *FULL* trials query the dependency between A and C. An example *FULL DIAGNOSIS* question in the *COMMON CAUSES* condition is “if a self-driving golf cart crashed, could it have been infected by malware?”

Simulation We simulate the behaviour of the model on experimental trials. We run $n = 2000$ replicates of each condition ($N = 24000$). The noise parameter is set relatively low ($\sigma = 0.01$) to highlight qualitative differences between conditions.

To set up each run, we initialize the model with the relevant causal network topology. On each trial, we set the evidence/given-event to have an initial activation of 1 in working memory. All other events are set to an initial activation of 0 in working memory. Furthermore, on each trial, meta-cognitive weights are configured to match the relevant condition, as discussed in the Model section.

We measure retrieval correctness and response time for each experimental trial. The rate of correctness or *success-rate* in each condition can be interpreted as a measure of probabilistic certainty, assuming that agents calculate certainty by sampling many simulations. We measure response times by the number of retrieval events before termination, consistent with the assumption that each retrieval operation incurs temporal cost (Anderson et al., 2004). Thus, fewer retrievals indicate faster, more efficient processing.

Analytical Approach We analyze simulated response times across conditions using a $3 \times 2 \times 2$ between-subjects ANOVA. We use post-hoc pairwise comparisons with independent samples t-tests to examine response time differences within each Scenario $\times$ Length condition and chi-squared tests to compare success-rates between level combinations. However, since both the number of samples and response variability were under experimenter control, we focus on reporting effect sizes rather than statistical significance (η^2 for ANOVA, d for independent t-tests, and ϕ for χ^2 -tests).

We expect response times to be larger for structures with more causal relationships than structures with fewer. Importantly, we expect an interaction between scenario and direction: predictions should take longer in the *COMMON CAUSES* condition (branching effects), diagnoses should take longer in the *COMMON EFFECTS* condition (branching causes), and no difference should be observed between directions in the *CHAIN* condition. Furthermore, we only expect failures in common effect diagnosis as inhibition between competing causes may prevent the retrieval of a query event when the model traverses an alternative branch of the causal network.

Results

Simulated response-time results are summarized in Figure 3. A full factorial ANOVA reveals a small three-way interaction



Figure 3: Left: Mean number of retrieval events required to reach the query chunk across causal structure conditions, reasoning directions, and inferential lengths. Error bars represent standard errors. Red dotted line indicates the theoretical minimum number of events based on the underlying causal network. Right: Rate of correctness of full length simulations across conditions

among SCENARIO, DIRECTION, and LENGTH ($\eta^2 = .01$). Because this interaction is small and not central to our primary predictions, we focus on theoretically relevant lower-order effects. The analysis reveals a large SCENARIO $\times$ DIRECTION interaction ($\eta^2 = .37$), indicating that the effect of reasoning direction depends strongly on causal structure. The model also shows large main effects of SCENARIO ($\eta^2 = .15$) and LENGTH ($\eta^2 = .47$), consistent with the claim that traversing larger causal structures requires more retrieval operations. In contrast, the main effect of DIRECTION is negligible ($\eta^2 < .001$). Smaller SCENARIO $\times$ LENGTH ($\eta^2 = .04$) and DIRECTION $\times$ LENGTH ($\eta^2 = .02$) interactions are also observed, likely because the COMMON CAUSE and COMMON EFFECT conditions are not structurally symmetrical.

Post-hoc comparisons provide additional insight on the SCENARIO $\times$ DIRECTION interaction. For the COMMON CAUSE scenario, response times in PREDICTION exceed DIAGNOSIS in SHORT, $d = 1.28$, and FULL, $d = 2.06$. The asymmetry is stronger in FULL likely because there are more branches available for traversal in FULL than SHORT, increasing possible traversal paths in PREDICTION. However, in the COMMON EFFECT scenario, DIAGNOSIS response times exceed PREDICTION response times in both the SHORT ($d = -1.36$) and FULL ($d = -1.24$) conditions.

Finally, success rate (i.e., *certainty*) is at ceiling in PREDICTION (100%) across all conditions. Similarly, success rate is at ceiling in all DIAGNOSIS trials except in the COMMON EFFECT condition (79.1%). These failures are likely caused by the model traversing a path that includes an event that shares an effect with the query event, preventing retrieval of the latter via lateral inhibition. Interestingly, in COMMON EFFECT DIAGNOSIS, success rates are lower in the SHORT condition (75.05%) than in the FULL condition (83.10%; $\phi = .10$).

Discussion

In this study, we present BLISA, a novel mechanistic model of causal reasoning that can account for empirical asymmetries in diagnostic (backward) and predictive (forward) causal inference. Additionally, we propose a novel experimental paradigm to uncover testable empirical predictions that follow from our model.

When presented with a causal query, BLISA incrementally constructs a causal narrative by retrieving causes or effects of events in working memory based on its causal knowledge. Depending on the direction of causal inference, BLISA prioritizes different retrieval patterns, allowing it to capture various empirical asymmetries in a mechanistically meaningful manner. Such empirical asymmetries include findings that predictions seem more certain than diagnoses (Medin et al., 2003; Tversky and Kahneman, 1980), that people neglect alternative causes when reasoning from cause to effect (Fernbach et al., 2011), and that alternative effects are more available than alternative causes during retrieval (Kelley, 1973; Redelmeier and Shafir, 2023).

Our simulation study illustrates how these mechanisms generate distinct behavioural predictions across causal structures. In particular, BLISA predicts that branching effects primarily increase retrieval time in predictive inference, whereas branching causes increase retrieval time and reduce certainty in diagnostic inference. This pattern is a consequence of the model’s assumption that diagnostic reasoning is subject to lateral inhibition among causes that share effects. Thus, the model predicts that processing difficulty depends on the interaction between causal direction and network topology.

Related Work

BLISA is a process-oriented cognitive architectural model that performs diagnostic and predictive causal reasoning primarily through memory mechanisms. In that regard, it is similar

to the ACT-R models of Mehlhorn et al. (2011) in its use of spreading activation and competitive retrieval. However, in Mehlhorn et al., these mechanisms are deployed for the purposes of hypothesis generation in diagnostic reasoning rather than the elaboration of causal narratives.

Indeed, as mentioned above, BLISA's elaboration of specific causal narratives is an interesting point of similarity with accounts of causal reasoning based on mental model theory (e.g., Goldvarg and Johnson-Laird, 2001; Khemlani et al., 2014). The mental model approach is also notable for presenting a modal view of causal relations, which may be contrasted with probabilistic views of causal inference that are characteristic of prominent normative theories (e.g., Pearl, 1986). By contrast, our model is compatible with either view as the model's design only depends on the relational nature of causality.

Finally, the Mutation Sampler (MS; Davis and Rehder, 2020; Rehder and Davis, 2021) is a resource-rational model (see Griffiths et al., 2015) of causal reasoning that has many similarities with BLISA. Notably, MS generates causal judgments by stochastically sampling specific causal scenarios using Metropolis-Hastings Markov Chain Monte Carlo. Moreover, causal scenarios generated by the MS model evolve incrementally as each new proposal modifies the value instantiated by exactly one causal variable. Despite these similarities, the two models address different levels of analysis. Namely, BLISA sits closer to the algorithmic and implementational levels than does MS, as the former is explicitly defined in terms of elementary psychological and cognitive constructs.

Future Work

Broadly, there are two main directions for future work. On the mechanistic side, BLISA makes a number of simplifying assumptions which may be relaxed to account for a potentially broader range of causal reasoning phenomena. One salient direction in this line is to include variation in antecedent and consequent weights to capture causal strength judgments (see, e.g., Rottman and Hastie, 2014). On the empirical side, testing the model against human behaviour is an important and necessary next step. As of the time of writing, work is underway to run our proposed experiment with human participants.

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